Reg. No. | | | | | | | | | | |

[18MTDS202]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

MACHINE LEARNING

Time: 03 Hours **Maximum Marks: 100**

Instructions:
- **i. Answer FIVE full questions. Question No. 1 and 2 are compulsory.**
- **ii. Draw neat sketches wherever necessary.**
- **iii. Make suitable assumptions wherever necessary.**

1. a. Just like a human, we can train a machine also. What do you think, in how many ways we can train a machine? Explain with suitable examples. **10**

b. Define Machine Learning. How would you develop Machine Learning Model? **10**

2. a. What do you understand by flexible generalization of ordinary linear regression? Difference between Generalized and General Linear Model. **10**

b. Explain the type of GLM model with an example, in which the random component is specified by the Poisson distribution of the response variable which is a count. **10**

3. a. What do you understand by odds and odds ratio? Explain the role of odds and odds ratio in logistic regression. **10**

b. Define the following terms: **10**

i. Logestic regression

ii. Maximum Liklihood Estimation

OR

4. a. Explain the Hosmer-Lemeshow and explain a problem with HL test. **10**

b. Explain the following terms: **10**

i. Confusion Matrix

ii. Binomial Regression

5. a. What do you understand by Decesion tree? Write ID3 Algorithm and explain. **10**

b. Give a proof for Wald test for Logistic regression. **10**

6. a. What you understand by ROC curve. Explain it with a suitable example. **10**

b. Write a brief note on various Kernel models with an example. **10**

7. a. What do you understand by feature extraction and what are the different types of feature extraction methods. **10**

b. Describe SVM and it's working principle with a suitabale example. **10**

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8. a. Consider the following data set consisting of the scores of two variables on each of seven individuals: Apply K-Means algorithm and form two clusters. **10**

Subject	A	B
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5
Cluster 2	4, 5, 6, 7	(4.1, 5.4)

b. Define clustering and compare Hierarchical and Partitioning clustering. **10**